Integration — Lesson 18 Test: Substitution II — u-Substitution

10 questions · show your work in the box · every antiderivative needs "+ C" · check answers by differentiating when you can

Name: ______________________________ Date: ______________ Score: ______ / 10

1. For the integral $\int 2x(x^2+4)^6\,dx$, state what u should be and what du equals. (Do not integrate — just identify the substitution.) 1 pt

2. Compute $\int 2x(x^2-3)^4\,dx$ using u-substitution. Show all six steps, including the differentiation check. 1 pt

3. Compute $\int 3x^2(x^3+5)^3\,dx$. 1 pt

4. Compute $\int \cos x (\sin x)^4 \, dx$. 1 pt

5. Compute $\int x(x^2+1)^3 \, dx$. (Watch out: du wants 2x, but the integral only gives you x.) 1 pt

6. Compute $\int x^2(x^3-4)^3 \, dx$. 1 pt

7. Does u-substitution work on $\int (x^3+2)^5 \, dx$? Explain why or why not, and say what method you would use instead. (You do not need to finish the computation.) 1 pt

8. A classmate claims that $\int x(x^2+7)^2\,dx = (x^2+7)^3/6 + C$. Differentiate the claimed answer to decide whether the classmate is right. Show the chain-rule steps. 1 pt

9. Compute the definite integral $\int_0^1 4x(x^2+1)\,dx$. (Find the antiderivative by u-substitution first, substitute back, then plug in the limits.) 1 pt

10. Water flows into a tank at a rate of $r(t) = 2t(t^2+1)$ liters per minute, where t is in minutes. How many liters flow in between $t = 0$ and $t = 2$? Use u-substitution and include units in your answer. 1 pt